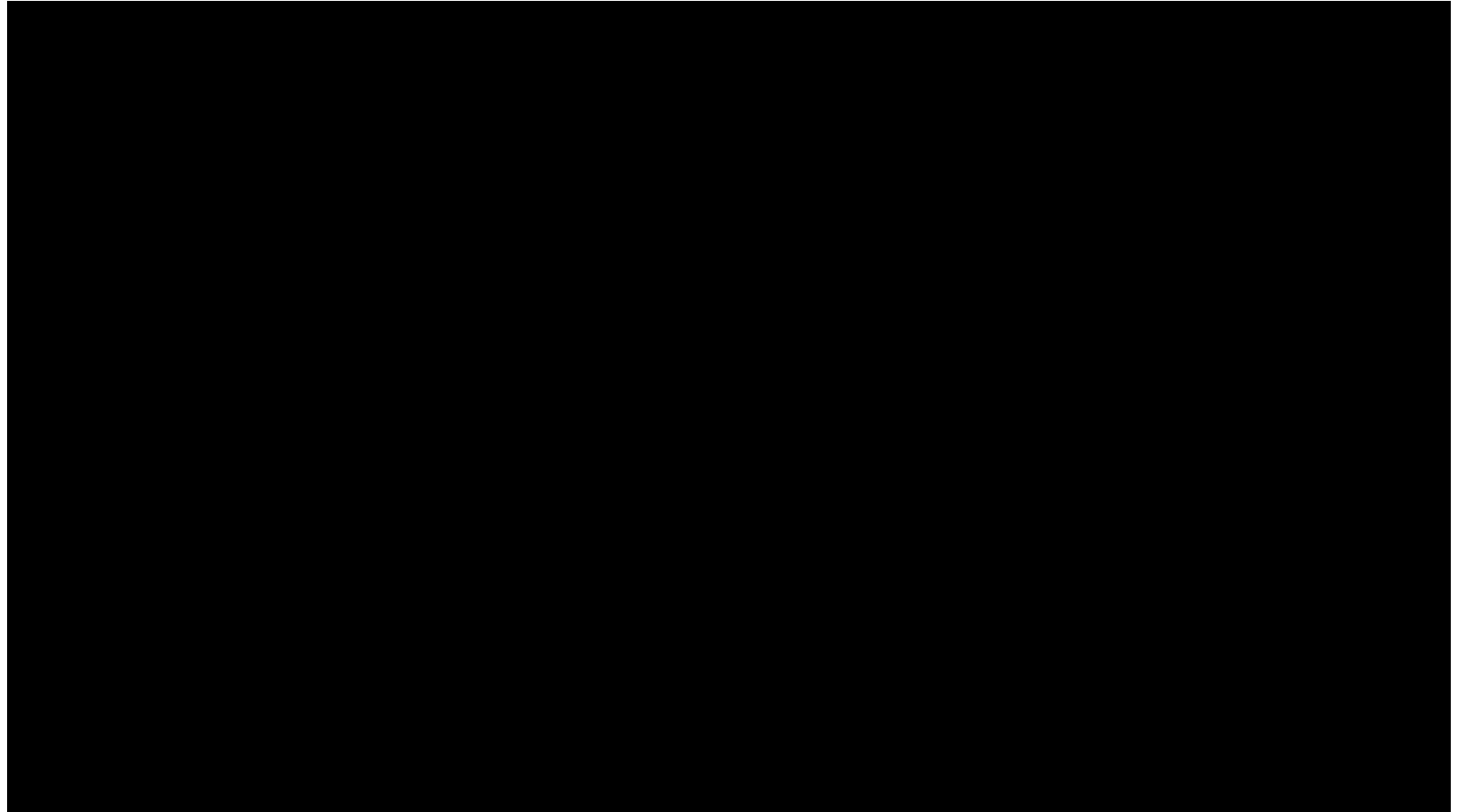


Lecture 1

Introduction

The Exciting, Perilous Journey Toward AGI | Ilya Sutskever



The 3W of Machine Learning

- **W**hat is Machine Learning (ML)?

What is **machine** ?

What is **learning**?

What is **machine learning**?

- **W**hy do we need Machine Learning?

The **necessity and importance** for machine learning

- **HoW** to perform Machine Learning?

The **building blocks** of a machine learning system

What is Machine Learning?

An example.....



What Can Machine Learning Do?

Some examples.....



Deep Blue



Unmanned Air Vehicle



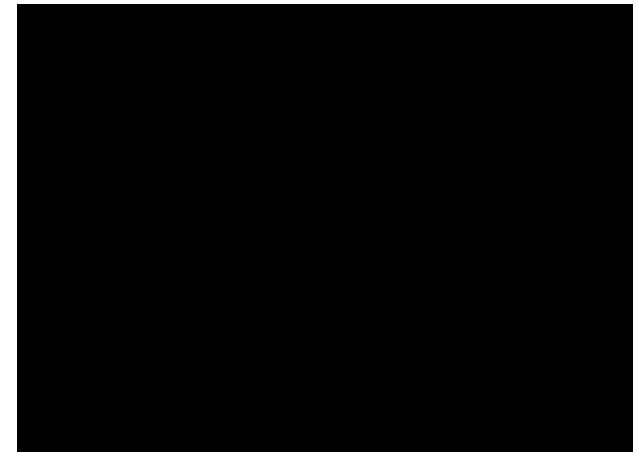
Autonomous Driving

```
daly@spiff: ~  
daly@spiff:~$ export AXIOM=/space/lambda/ent/ubuntu  
daly@spiff:~$ export PATH=$AXIOM/bin:$PATH  
daly@spiff:~$ axiom -nox  
AXIOM Computer Algebra System  
Version: Axiom (July 2009)  
Timestamp: Sunday August 9, 2009 at 15:26:54  
Issue |copyright to view copyright notices.  
Issue |summary for a summary of useful system commands.  
Issue |quit to leave AXIOM and return to shell.  
-----  
Re-reading compress.daae Re-reading interp.daae  
Re-reading operation.daae  
Re-reading category.daae  
Re-reading browse.daae  
(1) -> f:=x/sqrt(x^4+10*x^2-96*x-71)  
(1) -----  
      4      2  
      \|x  + 10x  - 96x - 71  
Type: Expression Integer  
(2) -> integrate(f,x)  
(2) -----  
      log  
      6      4      3      2      4      2      8  
      (x + 15x  - 80x  + 27x  - 528x + 781)\|x  + 10x  - 96x - 71 - x  
      +  
      6      5      4      3      2      8  
      - 20x  + 128x  - 54x  + 1408x  - 3124x  - 10001  
      /  
      8  
(3) ->  $\square$   
Type: Union(Expression Integer,...)
```

Symbolic Computation



Machine Translation



Digit Recognition

Definition by ChatGPT

SU

什么是机器学习？



机器学习是一种人工智能的分支，它使用算法和统计模型使计算机系统能够从数据中自动学习，而不需要显式地编程。机器学习的目标是通过让计算机系统从数据中学习来改进其性能和准确性，以便它们可以自动完成复杂的任务。



机器学习的基本原理是从历史数据中发现模式和规律，以便在未来的新数据上进行预测或决策。这种学习可以通过各种算法和技术来实现，例如神经网络、决策树、支持向量机和贝叶斯分类器等。

机器学习应用广泛，包括语音识别、图像识别、自然语言处理、推荐系统、金融风险管理等领域。它已经成为当今人工智能发展的重要组成部分，具有广泛的应用前景和深远的影响。

Note that the output of ChatGPT may vary with each interaction

What Can Machine Learning Do?

ChatGPT (OpenAI)

USER what is the fermat's little

CHATGPT Fermat's Little Theorem states that if a is not divisible by p , then a^p (a to the power of p) is congruent to a modulo p . In other words, if a is not divisible by p , then a^p divided by p has the same remainder as a .

For example, if $p = 5$ and $a = 2$, then $2^5 = 32$ is congruent to 2 modulo 5 . Therefore, 2^5 is congruent to 2 modulo 5 in this case.

Fermat's Little Theorem is necessary to perform modular arithmetic, which is a useful tool for proving many results in number theory.



; Given an array of integers nums and an integer target, return indices of the two numbers such that they add up to target. You may assume that each input would have exactly one solution, and you may not use the same element twice. You can return the answer in any order.

```
section .data
nums dd 1, 2, 3, 4, 5 ; array of integers
target dd 6 ; target integer
```

```
section .text
global _start
```

```
_start:
mov ebp, esp ; set up stack frame
push ebp
mov ebp, esp
```

Copy code

```
; initialize variables
mov eax, 0 ; first index
mov ebx, 0 ; second index
mov ecx, 0 ; sum of two numbers

; loop through nums array
mov edx, [nums] ; edx points to first element of nums
mov esi, 0 ; esi is index counter
```

What Can Machine Learning Do?

OpenAI Sora: Creating video from text



Prompt: A young man at his 20s is sitting on a piece of cloud in the sky, reading a book.

Generated by Minimax h3



Demo by Spatial Intelligence



Definition of (Machine) Learning



Herbert Simon
(1916—2001)

Learning is any process by which a system improves performance from experience.

Nobel Prize for Economics (1978)
Turing Award (1975)



Tom Mitchell

Founders University Professor at CMU

Machine Learning is the study of algorithms that

- improve their performance P
- at some task T
- with experience E .

Types of Machine Learning Algorithm

- **Supervised Learning**

- Given: training data + desired outputs (labels)

- **Semi-Supervised Learning**

- Given: training data + a few desired outputs

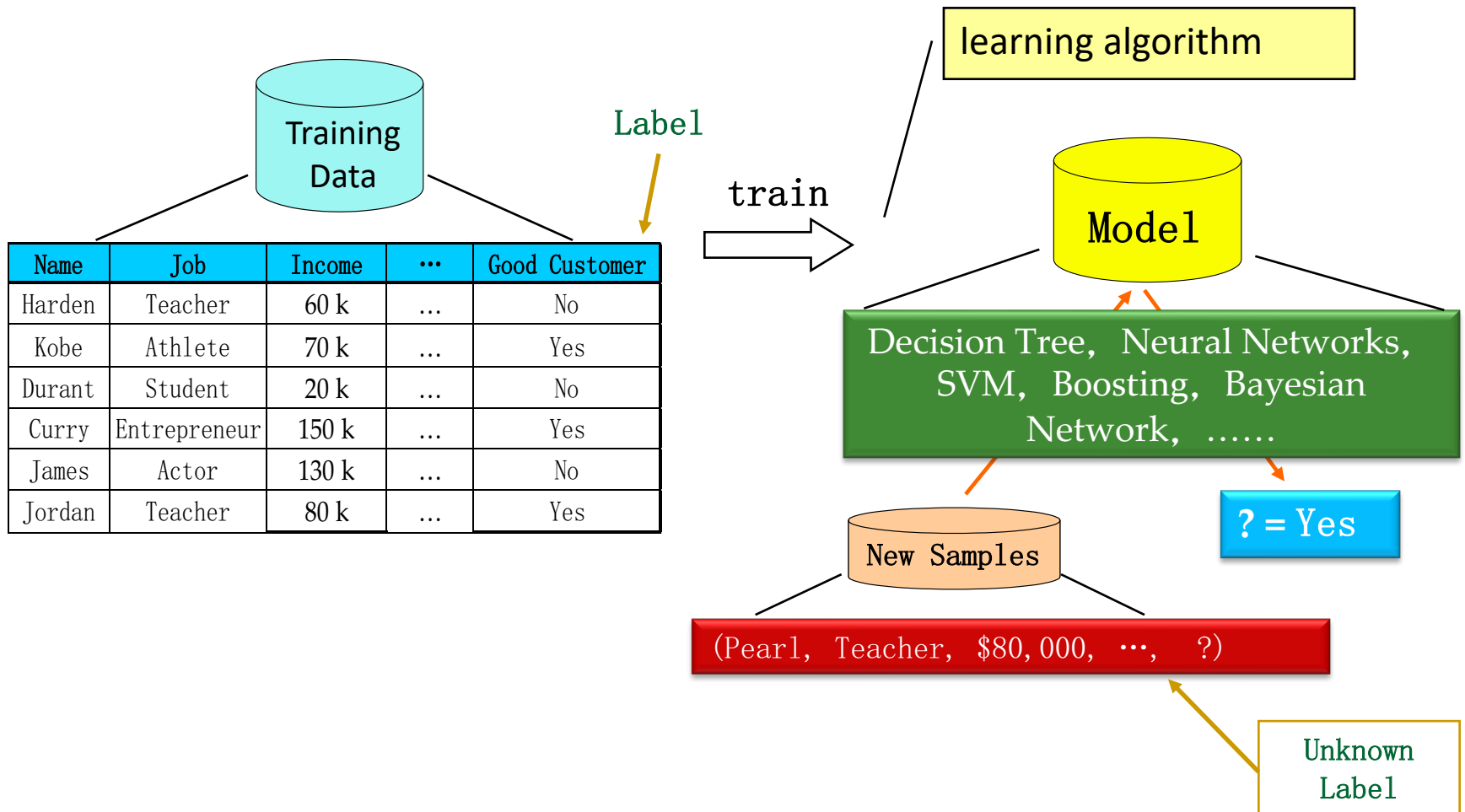
- **Unsupervised Learning**

- Given: training data (without desired outputs)

- **Reinforcement learning**

- Rewards from sequence of actions

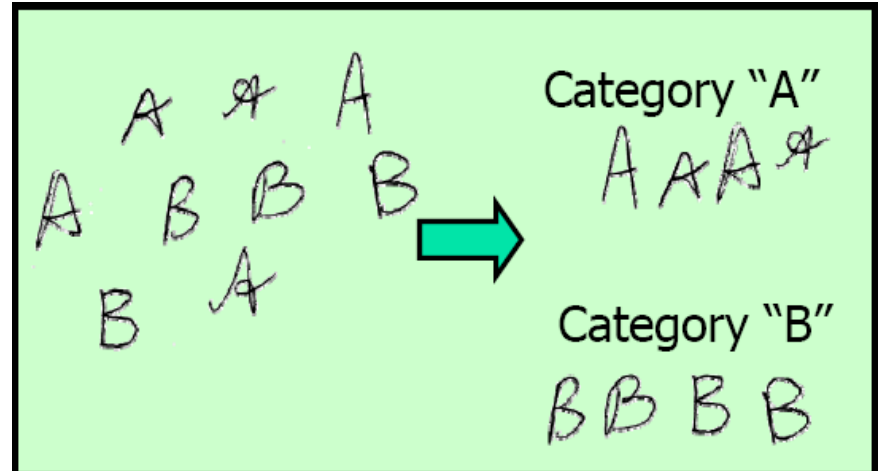
Supervised Learning



Supervised vs. Unsupervised Learning

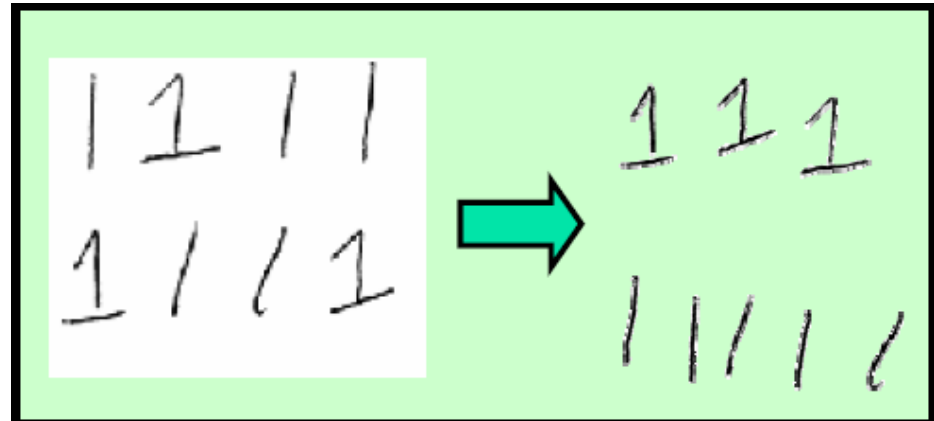
Classification: An example

We **already know** the categories of characters, and then classify the handwritten ones into category "A" and category "B"



Clustering: An example

We **do not know** the categories of symbols, and then learn the categories and group the symbols accordingly



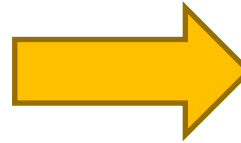
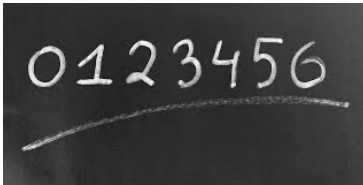
Applications of Machine Learning

1) Character Recognition

[字符识别]

Input:

images with characters



Output:

the identified
character strings
(0123456)

Useful in scenarios such as **automatic license plate recognition (ALPR)**, **optical character recognition (OCR)**, etc.

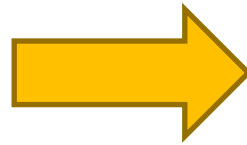
Applications of Machine Learning (Cont.)

2) Speech Recognition

[语音识别]

Input:

acoustic signal
(e.g. sound waves)



Output:

contents of the speech



“Imagine all the people.
Living life in peace”
——John Lennon

Applications of Machine Learning (Cont.)

3) Face Detection [人脸检测]

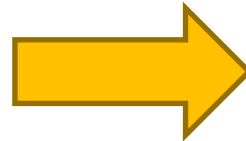
Useful in scenarios such as **digital camera capturing, video surveillance**, etc.

Input:

images with several people

Output:

locations of the peoples' faces in the image



Applications of Machine Learning (Cont.)

4) Text Categorization

[文档分类]

Input:

document, web pages, etc.



A general view shows the Independence Monument in central Kyiv, Ukraine February 25, 2022. REUTERS/Valentyn Ogrenko

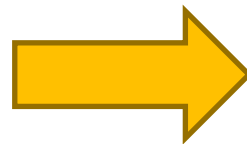
KYIV, Feb 26 (Reuters) - Ukraine on Saturday denied suggestions that it was refusing to negotiate a ceasefire with Russia but said it was also not ready to accept ultimatums or unacceptable conditions.

Mykhailo Podolyak, an adviser to the office of Ukrainian President Volodymyr Zelenskiy, told Reuters Ukraine has prepared a negotiating position but was faced with impractical negotiating conditions from Russia.

"It was yesterday that the aggressive actions of the armed forces of the Russian Federation escalated, up to evening and night mass air and missile strikes on Ukrainian cities," he said in a message. "We consider such actions only an attempt to break Ukraine and force it to accept categorically unacceptable conditions."

Reporting by Pavel Polityuk; writing by Matthias Williams; Editing by Catherine Evans

Our Standards: [The Thomson Reuters Trust Principles.](#)



Output:

category of the text, such as political, economic, military, sports, etc.

Useful in scenarios such as
information retrieval,
document organization, etc.

How to perform Machine Learning?

For **humans**, learning is **natural & easy**

recognize a face

understand spoken words

read handwritten characters

identify items by feel

decide whether an apple is ripe by its smell

For **computers**, learning is **never easy**

All in all, machine learning is **important, useful, attractive, but rather challenging**

Challenges → Opportunities

Basic Concepts

Model (模型)

Descriptions which are typically mathematical in form
[以数学形式表达的性质]

e.g. image → matrix; sound waves → frequency vector

Sample (样本)

Representatives of the patterns we want to classify
[分类的基本对象，学习的实例]

e.g. fingerprint of a suspect; Electrocardiogram of a patient

Training Set (训练集)

A set of samples used to train classifiers
[用于训练模型的样本集合]

Basic Concepts (Cont.)

Test Set (测试集)

A set of samples to be classified, **usually being mutually exclusive to training set**

[用于测试分类器的样本集合,通常与训练集无交集]

“Training set” vs. “Test set” $\Leftrightarrow$ “Homeworks” vs. “Exams”

Feature (特征)

Attributes which characterize properties of the samples

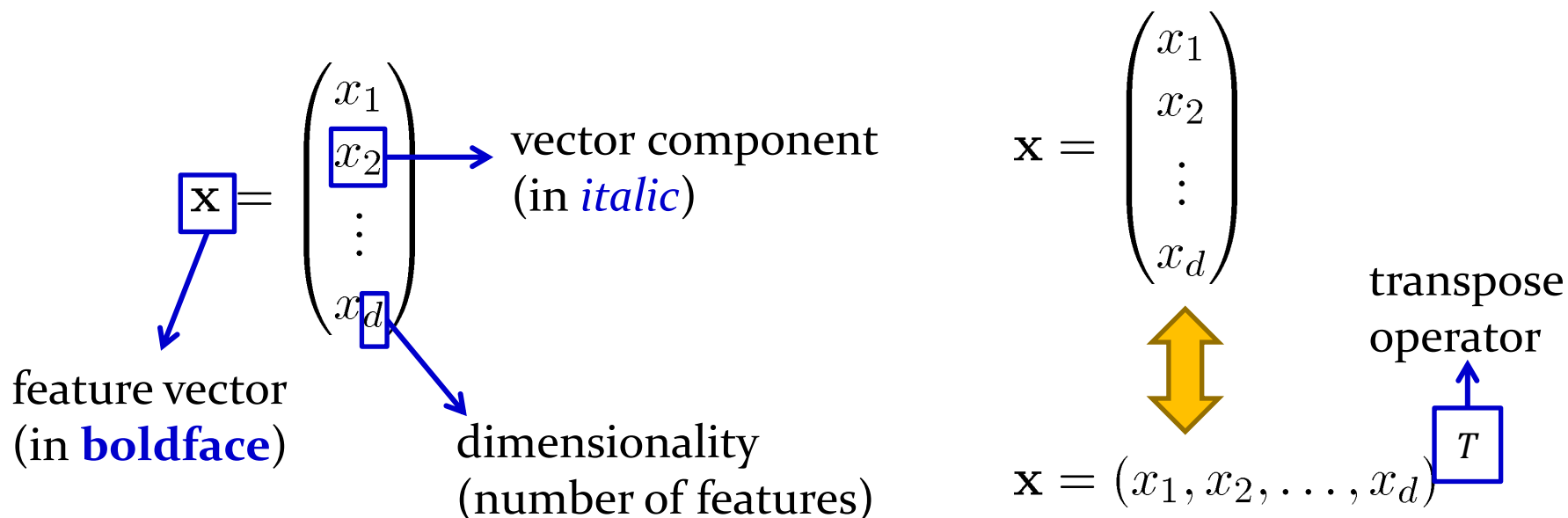
[用于刻画样本性质的属性]

e.g. to characterize a person, we may use features such as height, weight, age, salary, occupation, etc.

Basic Concepts (Cont.)

Feature Vector (特征向量)

Vector formed by a group of features, usually in column form
[由一组特征组成的向量，通常表示为列向量]



Basic Concepts (Cont.)

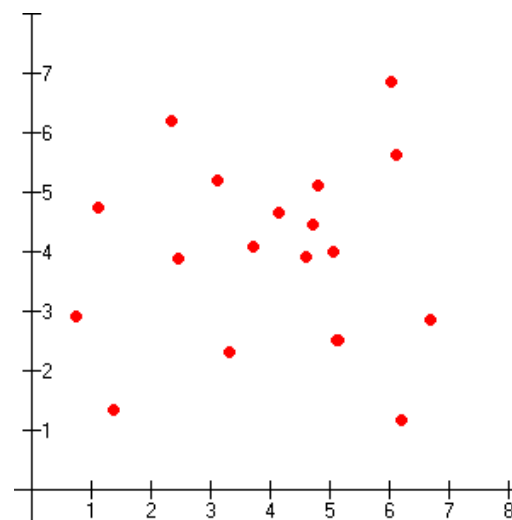
Feature Space (特征空间)

Space containing all the possible feature vectors
(由所有可能的特征向量组成的数据空间)

e.g. the d -dimensional Euclidean space $\mathbb{R}^d$

Scatter Plot (散布图)

Each sample is plotted as a
point in the feature space
(将每个样本表示为特征空间
中的一个点)

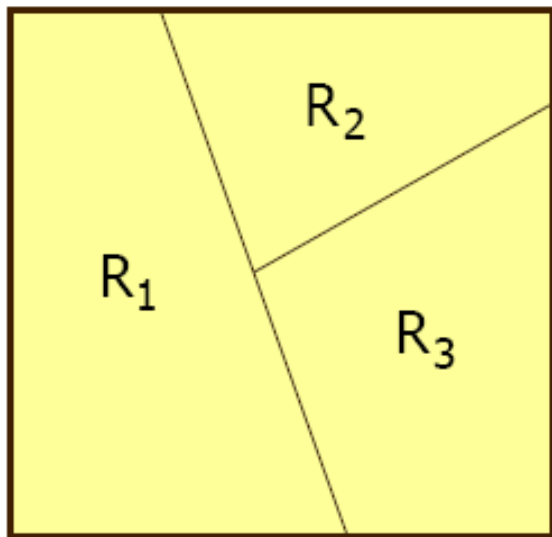


a 2D scatter plot

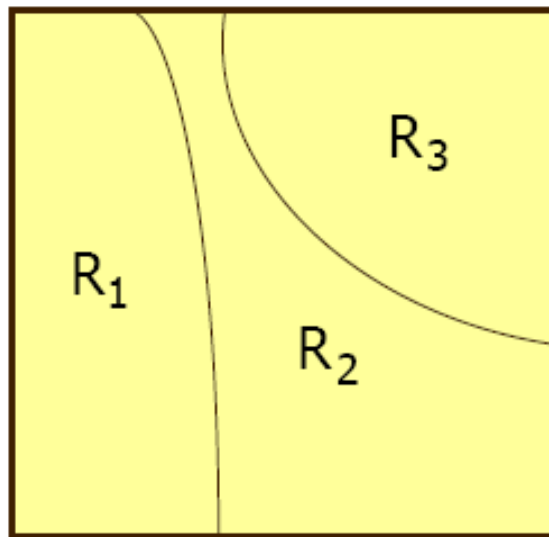
Basic Concepts (Cont.)

Decision Boundary (决策边界)

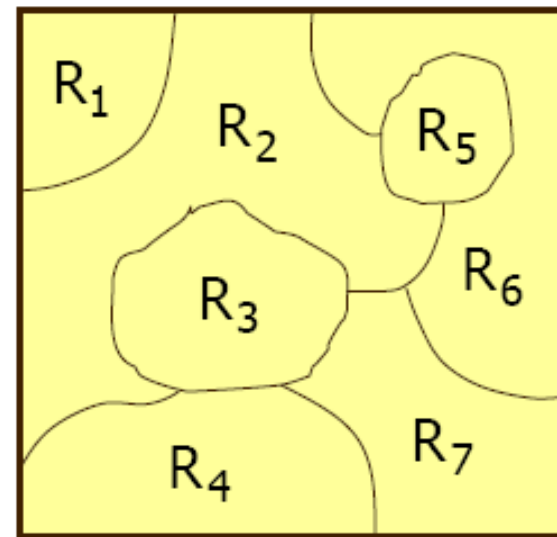
Boundaries in feature space which separate different categories
(特征空间中区分各个类别的边界)



linear boundary



quadratic boundary

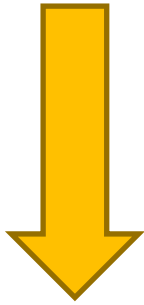


complex boundary

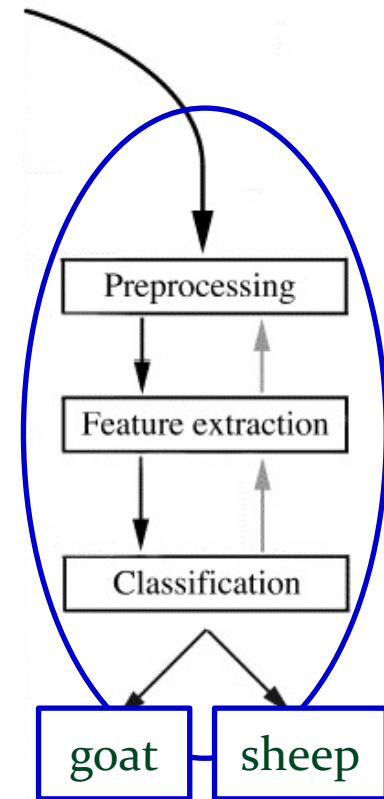
How to do Machine Learning?

An Example

The task: Automate the process of **separating animals** according to species



Separate *goat* from *sheep*
[山羊 vs. 绵羊]



Three basic steps

Example: “goat” vs. “sheep” (Cont.)

Step I: Preprocessing (预处理)

Goal: Preprocess the image captured by the camera, such that subsequent operations could be simplified without losing relevant information

Routine image
processing



- ☐ Adjust the level of illumination
- ☐ Denoising
- ☐ Enhance the level of contrast

segmentation



- ☐ Isolate different objects from one another
- ☐ Isolate objects from the background

.....

Example: “goat” vs. “sheep” (Cont.)

Step II: Feature Extraction (特征抽取)

Goal: Extract features (with good distinguishing ability) from the preprocessed image to be used for subsequent classification

Goat is usually
larger than sheep



“**size**” could be a good candidate
for features

Sheep wool is
usually **more**
curly than goat



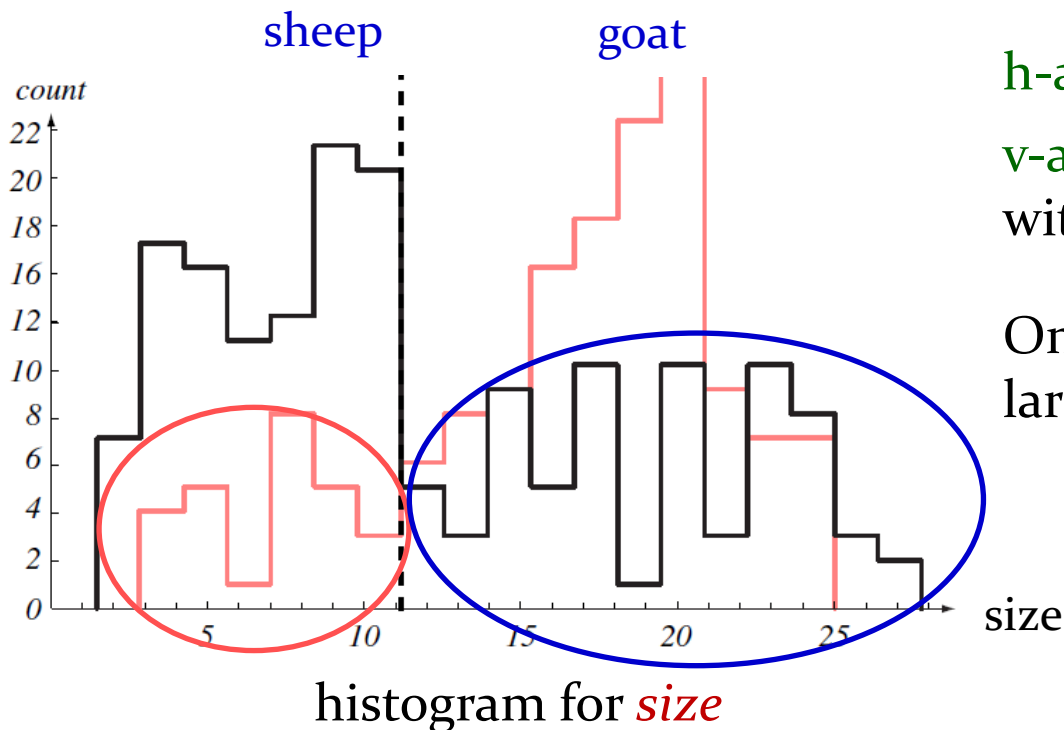
“**curliness**” could be another good
candidate for features

.....

Example: “Goat” vs. “Sheep” (Cont.)

Step III: Classification (分类)

Goal: To distinguish different types of objects (in this case, *Goat* vs. *Sheep*) based on the extracted features



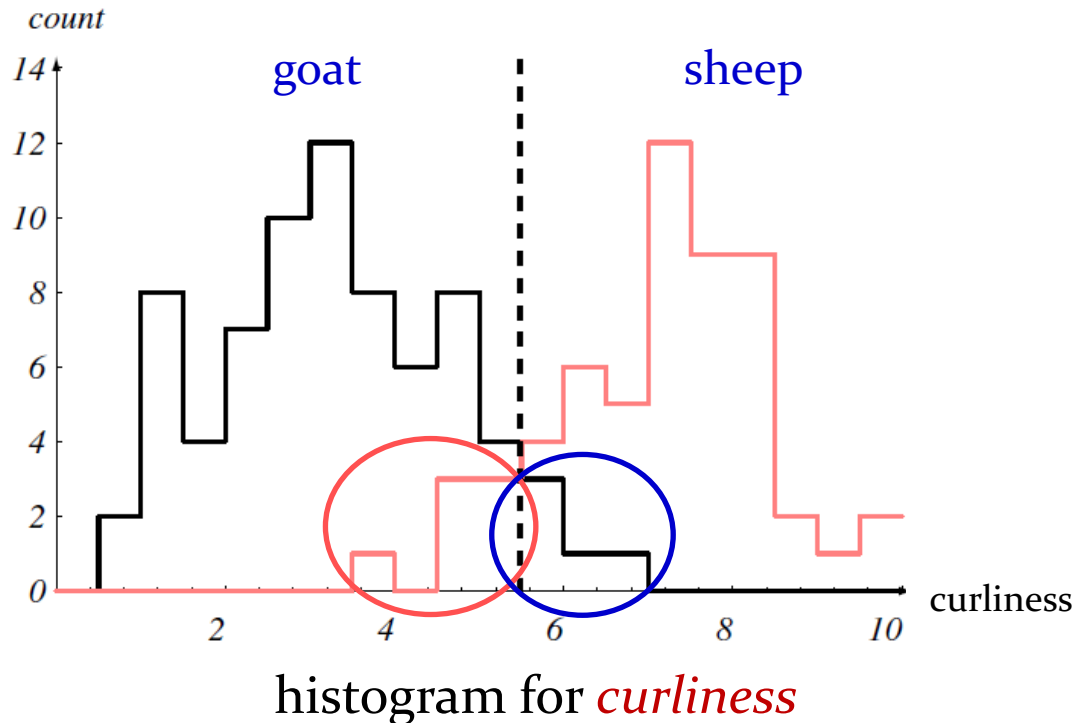
h-axis: size of the animals

v-axis: number of animals with a certain size

On average, goat is **somewhat** larger than sheep

Too much overlaps → poor separation with the length feature

Example: “Goat” vs. “Sheep” (Cont.)



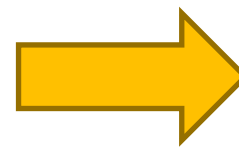
h-axis: curliness of wool

v-axis: number of animals with a certain curliness

On average, sheep wool is **much** curlier than goat

Less overlaps → better separation with the lightness feature, but still a bit unsatisfactory

What if no other single feature yields better performance?

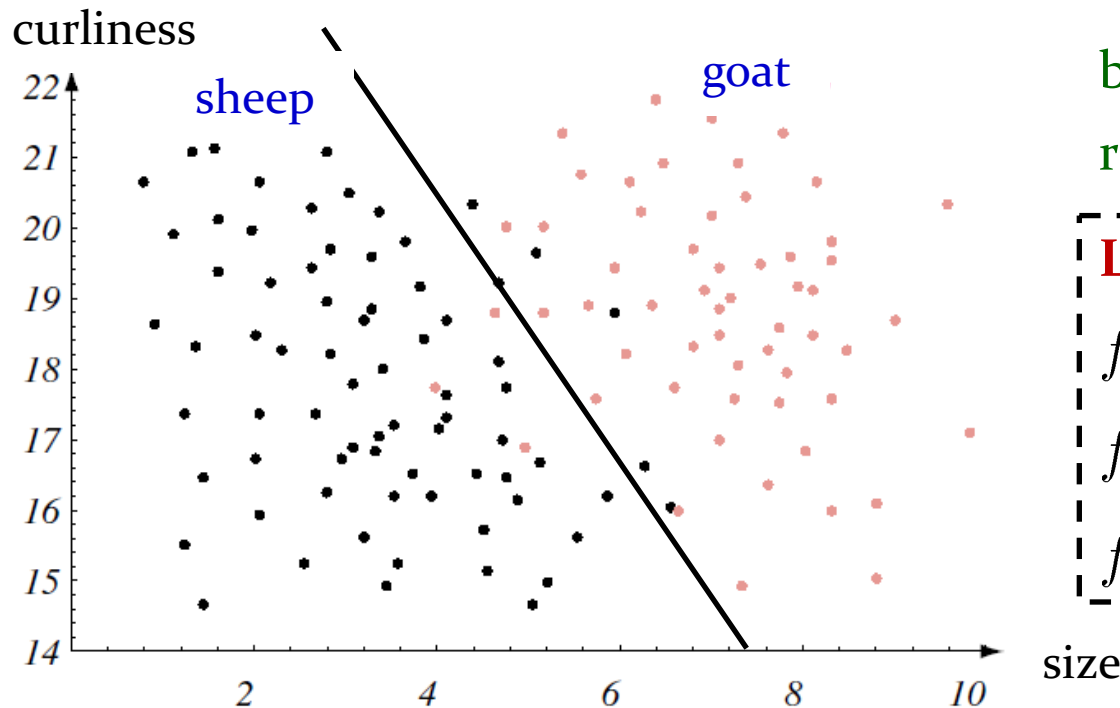


Use more features at the same time!

Example: “Goat” vs. “Sheep” (Cont.)

Using two features
simultaneously

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$



scatter plot for the feature vectors

black dots: sheep samples

red dots: goat samples

Linear decision boundary:

$$f(x_1, x_2) = a \cdot x_1 + b \cdot x_2 + c$$

$$f(x_1, x_2) > 0 \implies \text{sheep}$$

$$f(x_1, x_2) \leq 0 \implies \text{goat}$$

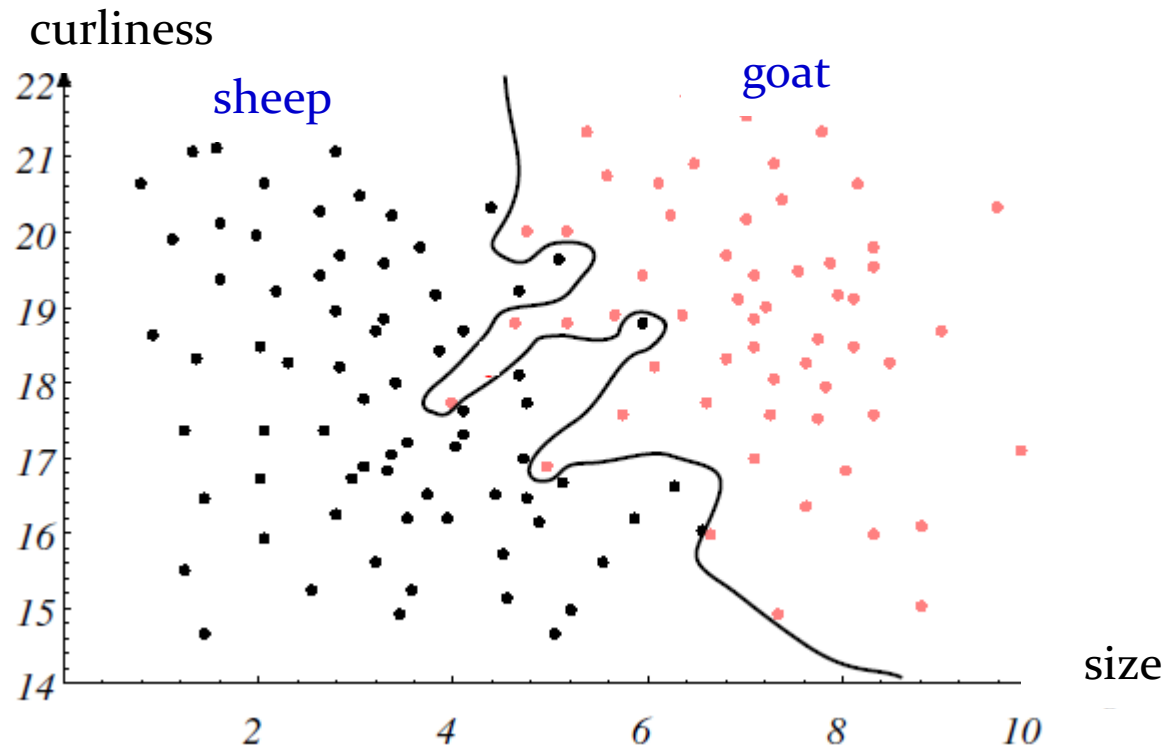
Much better than
single feature

Example: “Goat” vs. “Sheep” (Cont.)

Linear decision boundary:



Complex decision boundary



All the **training samples** (i.e. known patterns) have been separated perfectly

Linear vs non-linear,
which one is better?

Example: “Sheep” vs. “Goat” (Cont.)

Generalization

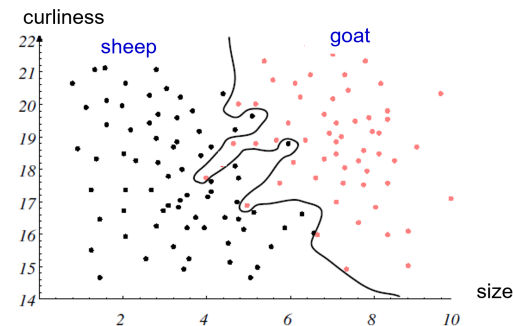
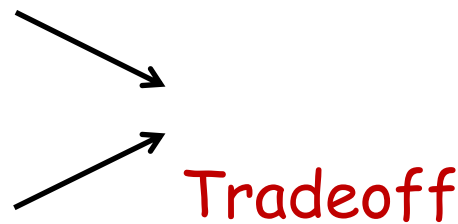
[泛化能力/推广能力]

The ultimate goal!

The central aim of designing a classifier is to **make correct decisions when presented with *novel (unseen/test)* patterns**, not on training patterns whose labels are already known

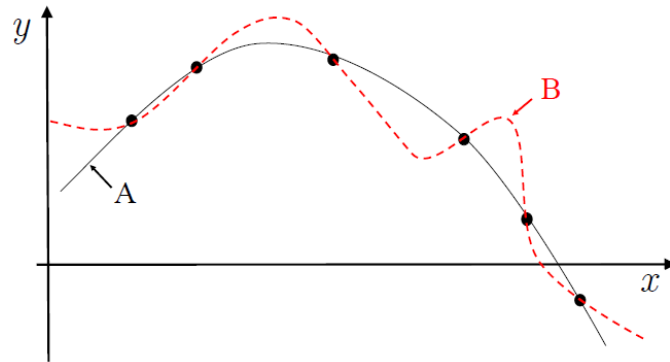
Performance on
the training set

Simplicity of
the classifier



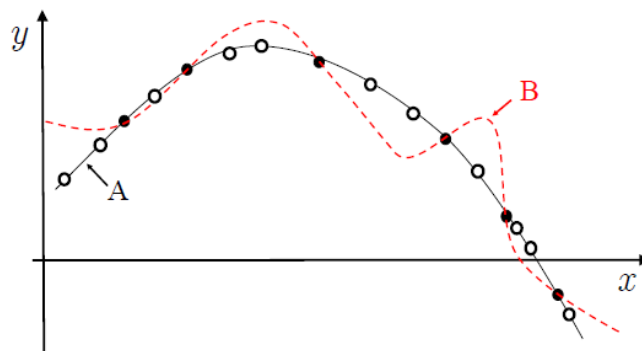
Inductive Bias

The bias of a learning algorithm towards a particular class of hypotheses is called the inductive bias.

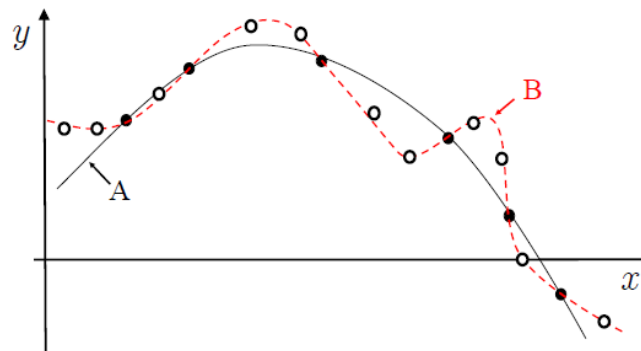


A or B?

There are infinite curves that are consistent with the finite training set.



(a) A is better.



(b) B is better.

Fig. 1.4: There is no free lunch. (● are training samples; ○ are testing samples)

Inductive Bias

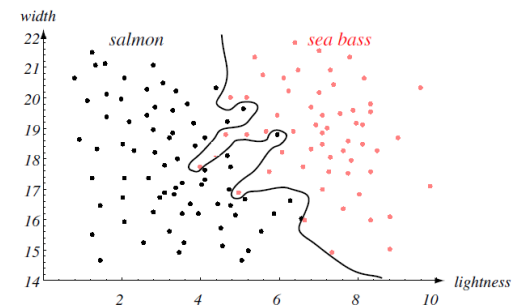
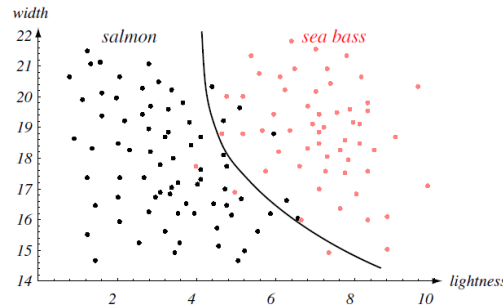
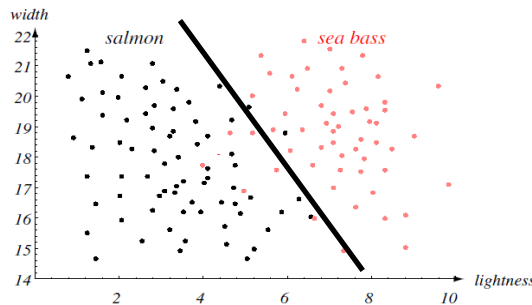
- We can interpret inductive bias as the heuristic or value philosophy of learning algorithms for search in potentially huge *hypothesis space*.
- A fundamental and widely used principle for this question in natural science is the *Occam's razor principle*, which says that we should choose the simplest hypothesis when there is more than one hypothesis consistent with the observations.
- In practice, whether this hypothesis matches the specific problem or not usually determines the performance of the model.

Model Selection

- Each machine learning method employs certain *model hypothesis*
- Every machine learning problem has its own *underlying true model*
- Fundamental questions on model selection
 - How do we know whether the hypothesized model is (relatively) consistent with the underlying true model?
 - How are we to know to reject a class of models and try another one?
 - Can we automate the process of model selection, instead of **trial and error** which is random and tedious?

Overfitting

- We can get perfect classification performance on the training data by choosing complex models
 - Complex models are **tuned to the particular training samples**, rather than the characteristics of the true model
- Models overly complex than necessary lead to overfitting
 - Good performance on the training data, but poor performance on novel data
- How can we find principled ways to obtain best complexity?



Classifier Ensemble

- Classifier ensemble aims to improve generalization performance by **employing a number of classifiers** for the same task
 - To improve the performance of speech recognizer: combine the results of *acoustic recognition* and *lip reading*
 - a.k.a. *Multi-classifier System*, *Mixture of Experts*, *Classifier Fusion*, etc.
 - Diverse ensemble techniques: *Bagging*, *Boosting*, *Random subspace*.
- How to combine different classifiers?
 - **Majority voting**: vote for the category where most classifiers agree
 - **Weighted voting**: weight each vote by classifier's confidence
 - **Stacking**: learn the rule of combination (more complicated)

Costs and Risks

- Cost is the loss after making incorrect decisions
 - **Equal cost:** In OCR, the cost of mistaking “6” as “9” might be **equal to** that of mistaking “9” as “6”
 - **Unequal cost:** In COVID-19 diagnosis, the cost of mistaking “positive (阳性)” as “negative (阴性)” would be **much higher than** that of mistaking “negative” as “positive” (**cost-sensitive learning**)
- Risk is total expected cost which we want to optimize
 - Error rate (percentages of test patterns being wrongly classified)
 - Precision, Recall, Area under the ROC curve (AUC), etc.
- Questions on costs and risks
 - How do we incorporate knowledge of costs, e.g. unequal cost?
 - Can we estimate the *lowest* possible risk of any classifier?
 -

No Free Lunch

The No Free Lunch (NFL) theorem

If a learning algorithm H_a outperforms another learning algorithm H_b in some situations, then H_b will outperform H_a in some other situations.

All learning algorithms are equally good considering all contexts. Debating “**which learning algorithm is better**” is meaningless without considering the specific task,

No Free Lunch

In binary classification problems, the target function could be any functions with a function space of $\mathcal{X} \mapsto \{0, 1\}$ with a function space of $\{0, 1\}^{|\mathcal{X}|}$. Summing the errors of f with respect to uniform distribution gives:

$$\begin{aligned}\sum_f E_{ote}(\mathcal{L}_a | X, f) &= \sum_f \sum_h \sum_{\mathbf{x} \in \mathcal{X}-X} P(\mathbf{x}) \mathbb{I}(h(\mathbf{x}) \neq f(\mathbf{x})) P(h | X, \mathcal{L}_a) \\ &= \sum_{\mathbf{x} \in \mathcal{X}-X} P(\mathbf{x}) \sum_h P(h | X, \mathcal{L}_a) \sum_f \mathbb{I}(h(\mathbf{x}) \neq f(\mathbf{x})) \\ &= \sum_{\mathbf{x} \in \mathcal{X}-X} P(\mathbf{x}) \sum_h P(h | X, \mathcal{L}_a) \frac{1}{2} 2^{|\mathcal{X}|} \\ &= \frac{1}{2} 2^{|\mathcal{X}|} \sum_{\mathbf{x} \in \mathcal{X}-X} P(\mathbf{x}) \sum_h P(h | X, \mathcal{L}_a) \\ &= 2^{|\mathcal{X}|-1} \sum_{\mathbf{x} \in \mathcal{X}-X} P(\mathbf{x}) \cdot 1.\end{aligned}$$

The sum of errors is independent of the learning algorithm !

- All learning algorithms are equally good considering all contexts.
- Debating “which learning algorithm is better” is meaningless without considering the specific task.

Generalization Error

Definitions of the **generalization error** and **empirical error** from “Foundations of Machine Learning”

Definition 2.1 (Generalization error) Given a hypothesis $h \in \mathcal{H}$, a target concept $c \in \mathcal{C}$, and an underlying distribution $\mathcal{D}$, the generalization error or risk of h is defined by

$$R(h) = \mathbb{P}_{x \sim \mathcal{D}} [h(x) \neq c(x)] = \mathbb{E}_{x \sim \mathcal{D}} [1_{h(x) \neq c(x)}],$$

where 1_ω is the indicator function of the event ω .²

The generalization error of a hypothesis is not directly accessible

Definition 2.2 (Empirical error) Given a hypothesis $h \in \mathcal{H}$, a target concept $c \in \mathcal{C}$, and a sample $S = (x_1, \dots, x_m)$, the empirical error or empirical risk of h is defined by

$$\hat{R}_S(h) = \frac{1}{m} \sum_{i=1}^m 1_{h(x_i) \neq c(x_i)}.$$

Generalization Error

For a fixed $h \in \mathcal{H}$, the expectation of the empirical error based on an *i.i.d.* sample S is equal to the generalization error: (independent and identically distributed)

$$\mathbb{E}_{S \sim \mathcal{D}^m} [\hat{R}_S(h)] = R(h).$$

Proof: by the linearity of the expectation and the fact that the sample is drawn *i.i.d.*, we can write

$$\mathbb{E}_{S \sim \mathcal{D}^m} [\hat{R}_S(h)] = \frac{1}{m} \sum_{i=1}^m \mathbb{E}_{S \sim \mathcal{D}^m} [1_{h(x_i) \neq c(x_i)}] = \frac{1}{m} \sum_{i=1}^m \mathbb{E}_{S \sim \mathcal{D}^m} [1_{h(x) \neq c(x)}],$$

for any x in sample S . Thus,

$$\mathbb{E}_{S \sim \mathcal{D}^m} [\hat{R}_S(h)] = \mathbb{E}_{S \sim \mathcal{D}^m} [1_{h(x) \neq c(x)}] = \mathbb{E}_{x \sim \mathcal{D}} [1_{h(x) \neq c(x)}] = R(h).$$

Generalization Error Bounds

The **generalization error** can be upper bounded by the **empirical error**

Corollary 3.19 (VC-dimension generalization bounds) *Let $\mathcal{H}$ be a family of functions taking values in $\{-1, +1\}$ with VC-dimension d . Then, for any $\delta > 0$, with probability at least $1 - \delta$, the following holds for all $h \in \mathcal{H}$:*

$$R(h) \leq \hat{R}_S(h) + \sqrt{\frac{2d \log \frac{em}{d}}{m}} + \sqrt{\frac{\log \frac{1}{\delta}}{2m}}.$$

Theorem 3.5 (Rademacher complexity bounds – binary classification) *Let $\mathcal{H}$ be a family of functions taking values in $\{-1, +1\}$ and let $\mathcal{D}$ be the distribution over the input space $\mathcal{X}$. Then, for any $\delta > 0$, with probability at least $1 - \delta$ over a sample S of size m drawn according to $\mathcal{D}$, each of the following holds for any $h \in \mathcal{H}$:*

$$R(h) \leq \hat{R}_S(h) + \mathfrak{R}_m(\mathcal{H}) + \sqrt{\frac{\log \frac{1}{\delta}}{2m}}$$

and $R(h) \leq \hat{R}_S(h) + \hat{\mathfrak{R}}_S(\mathcal{H}) + 3\sqrt{\frac{\log \frac{2}{\delta}}{2m}}.$

Machine Learning Conference & Journal

Machine Learning Journals

- **JMLR** 《Journal of Machine Learning Research》
- **TPAMI** 《IEEE Transactions on Pattern Analysis and Machine Intelligence》
- **TKDE** 《IEEE Transactions on Knowledge and Data Engineering》
- **TNNLS** 《IEEE Transactions on Neural Network and Learning Systems》
- 国内：《中国科学 信息科学》、FCS
- ...

Machine Learning Conferences

- **ICML** (International Conference on Machine Learning)
- **NeurIPS** (Neural Information Processing Systems)
- **ICLR** (International Conference on Learning Representations)
- **KDD** (ACM SIGKDD Conference on Knowledge Discovery and Data Mining)
- **AAAI** (AAAI conference on Artificial Intelligence)
- **IJCAI** (International Joint Conference on Artificial Intelligence)
- 国内: MLA/CCML/CCDM/CCFAI

Machine Learning History & Future

History of AI

Marvin Minsky



John McCarthy



Julian Bigelow



Claude Shannon



Allen Newell

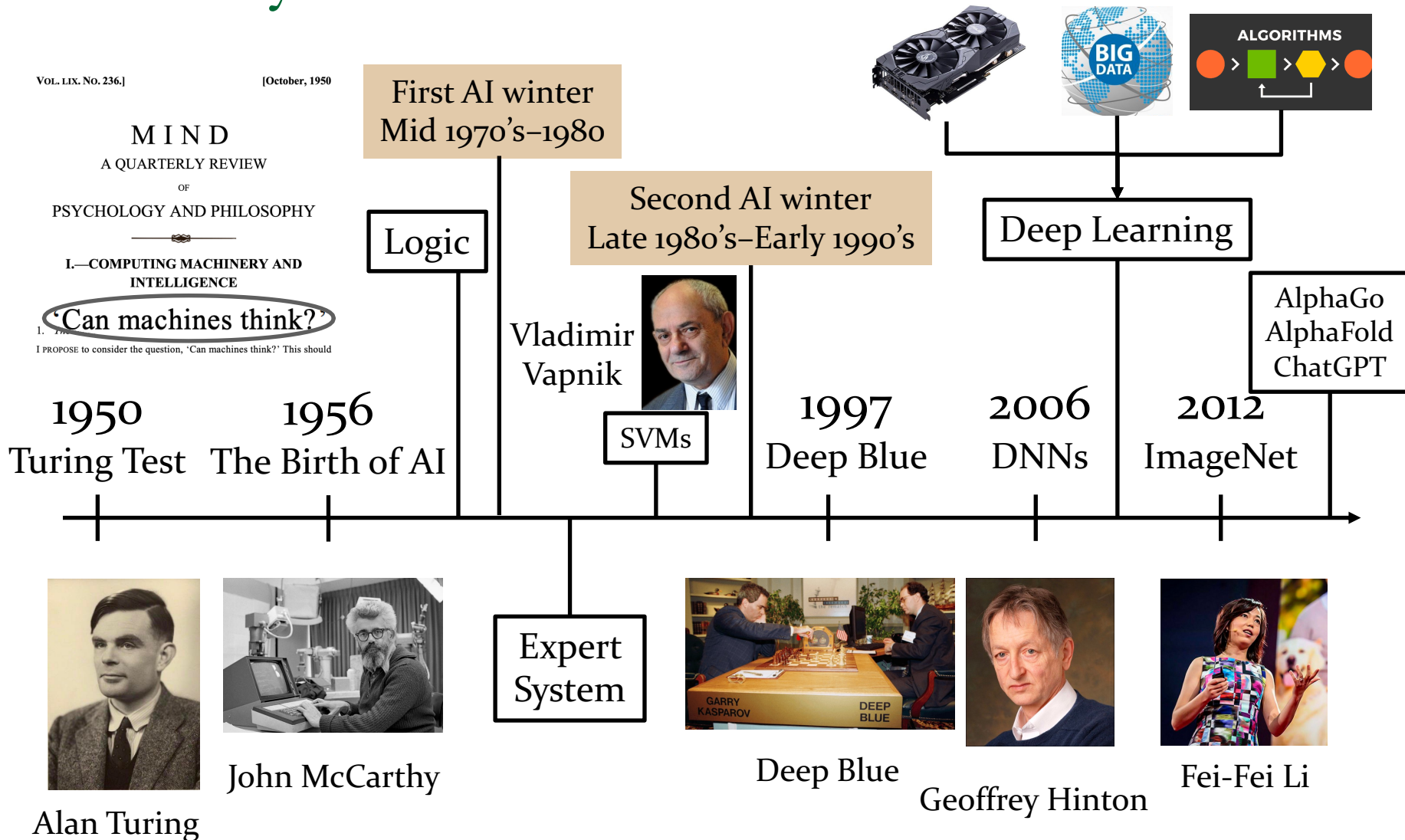


Herbert Simon



Dartmouth Workshop 1956

History of AI



The Future of AI



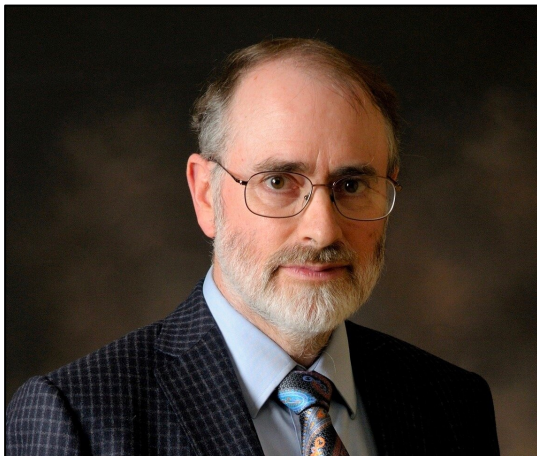
PROCEEDINGS OF THE IRE
Steps Toward Artificial Intelligence*
MARVIN MINSKY†, MEMBER, IRE

1961

Marvin Minsky (1927-2016)

人工智能先驱

2016



Tom Dietterich

AAAI/ACM Fellow

AAAI Chair (2016)

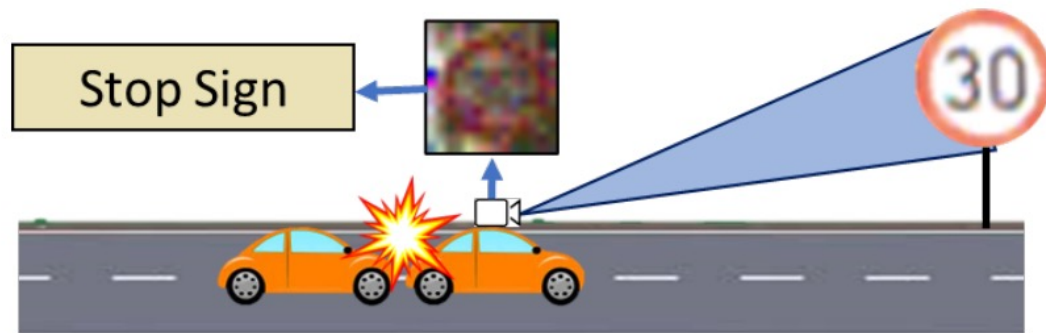
**STEPS TOWARD ROBUST
ARTIFICIAL INTELLIGENCE**

Tom Dietterich
President, Association for the Advancement of Artificial
Intelligence

Robust AI

■ The Need for Robust AI

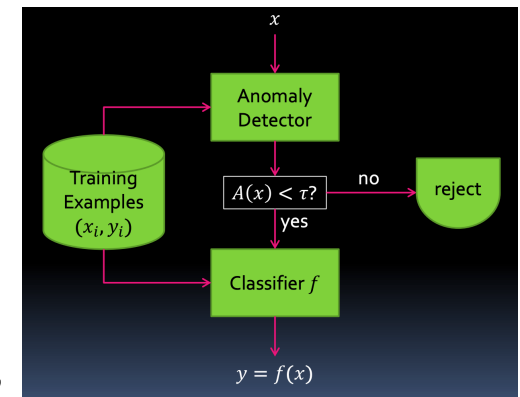
- High Stakes Applications
- Need to Act in the face of Unknown Unknowns



e.g., the detected image may suffer from various perturbations such as low resolution

■ Approaches toward Robust AI

- Robustness to Known Unknowns
- Robustness to Unknown Unknowns



An example: Anomaly Detection

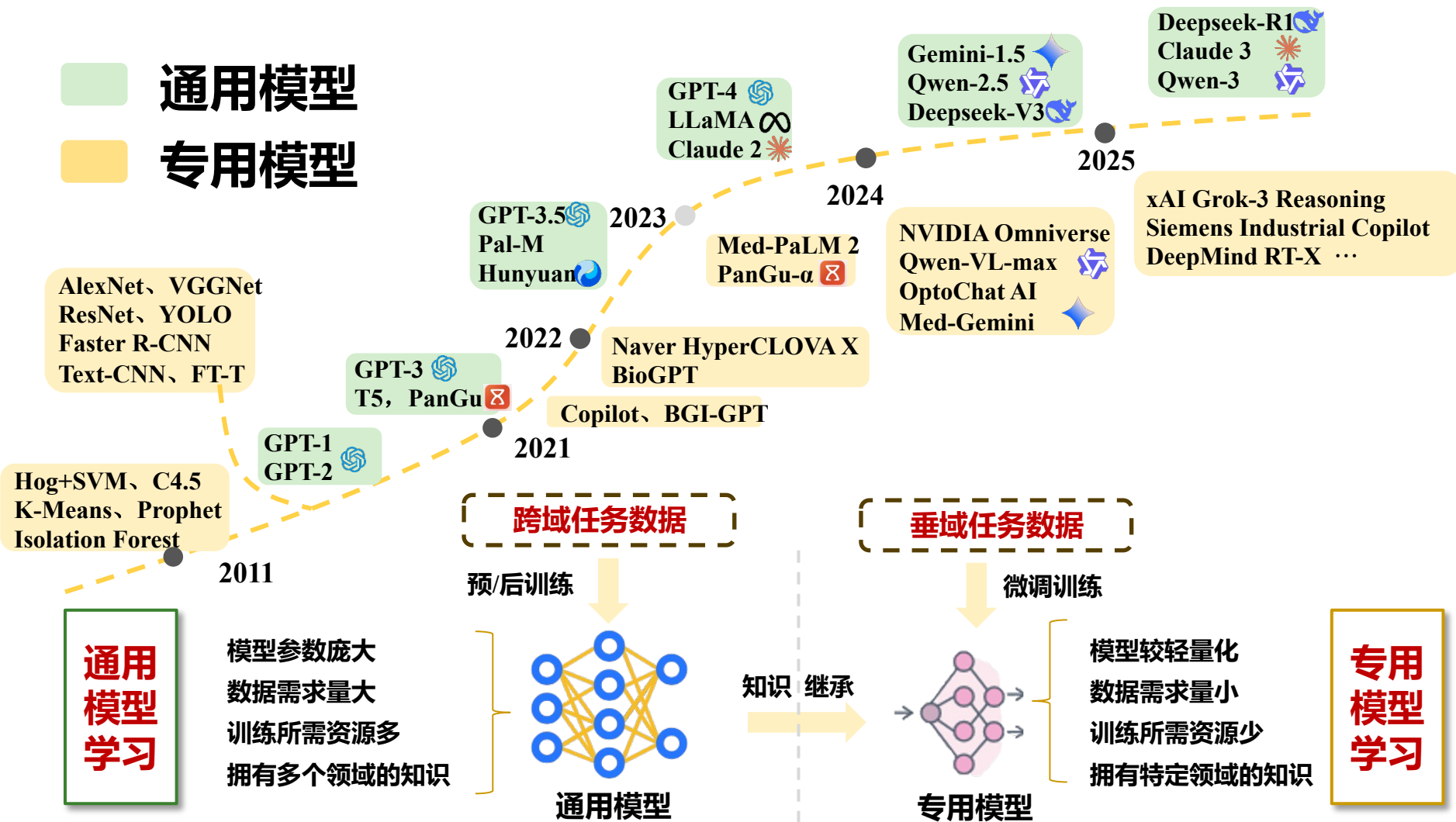
Minsky: Difference between Computer Program and Human

“almost any error will completely paralyze *a typical computer program*, whereas a person whose brain has failed at some attempt will find some other way to proceed. We rarely depend upon any one method. We usually know *several different ways* to do something, so that if one of them fails, there's always another.”

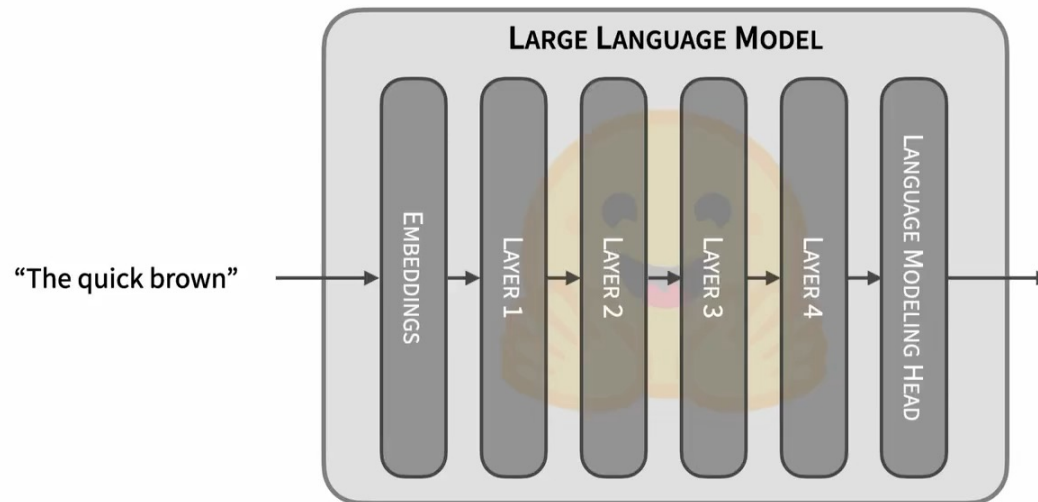
人工智能模型发展趋势

通用模型

专用模型



ChatGPT (OpenAI) - Text Generation



ChatGPT (OpenAI) - Text Generation

ChatGPT (OpenAI) - RLHF

RLHF: Reinforcement Learning from Human Feedback

Step 1

Collect demonstration data and train a supervised policy.

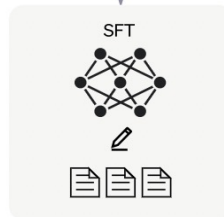
A prompt is sampled from our prompt dataset.



A labeler demonstrates the desired output behavior.



This data is used to fine-tune GPT-3.5 with supervised learning.



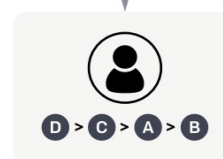
Step 2

Collect comparison data and train a reward model.

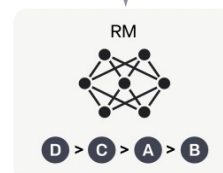
A prompt and several model outputs are sampled.



A labeler ranks the outputs from best to worst.



This data is used to train our reward model.



Step 3

Optimize a policy against the reward model using the PPO reinforcement learning algorithm. Proximal Policy Optimization

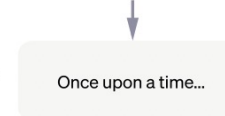
A new prompt is sampled from the dataset.



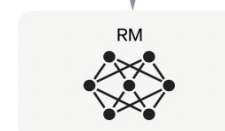
The PPO model is initialized from the supervised policy.



The policy generates an output.



The reward model calculates a reward for the output.



The reward is used to update the policy using PPO.



$$\text{GPT-4} \stackrel{?}{=} \text{UL} + \text{SL} + \text{RL}$$

The DeepSeek Series

2023.12

DeepSeek-V1

2024.5

DeepSeek-V2

- Their first MoE model with innovations like MLA; close to GPT-4
- Much lower pricing compared to other LLM providers
- Received attention from AI professionals

2024.12

DeepSeek-V3

- Beating GPT-4o
- Innovations led to low training costs (\$5.6 million)

Shared one pretrained model (DeepSeek-V3-Base)

2025.1

DeepSeek-R1

- Close to GPT-o1
- Went viral in mainstream media around the world

Only post-training part is different

The DeepSeek Series



- **Pre-training:** Learns language and knowledge via autoregressive modeling on unlabeled corpus.

$$\mathcal{L} = - \sum_{t=1}^T \log P(x_t | x_{<t}) \quad P(x_t | x_{<t}) = \text{Softmax}(\phi(x_{<t}))$$

- **Post-training:** Enables pre-trained models to follow instructions, answer questions, etc.
 - SFT: Train with cross-entropy loss on instruction-response data. – Teacher Forcing

$$\mathcal{L}_{\text{SFT}} = - \sum_{t=1}^T \log P_{\theta}(y_t | y_1, y_2, \dots, y_{t-1}, \mathbf{x})$$

- RL: Learn to generate more plausibly on (ranking or contrastive) preference data, which can be manually labeled or generated.

PAttern Learning and Mining Lab



[Introduction](#)

[News](#)

[Members](#)

[Academics](#)

[中文首页](#)

[Application](#)

PAttern Learning and Mining(PALM) Lab



Research@PALM

- Machine Learning & Data Mining
 - Multi-instance and multi-label learning
 - Semi-supervised and active learning
 - Cost-sensitive and class-imbalance learning
 - Structure learning and clustering
 - Label-distribution learning
 - Partial label learning
- Computer Vision (e.g., VLM)
- Natural Language Processing (e.g., LLM)